

nodes. Network depth and width are then selected through the hyperparameter search. In the tuned configuration, summarised in Table 5.1, the model grows into a deep dense architecture with nine hidden/output stages and a final linear regression output. The resulting structure remains conceptually feedforward, while its concrete layer widths are adapted to the ageing-estimation task.

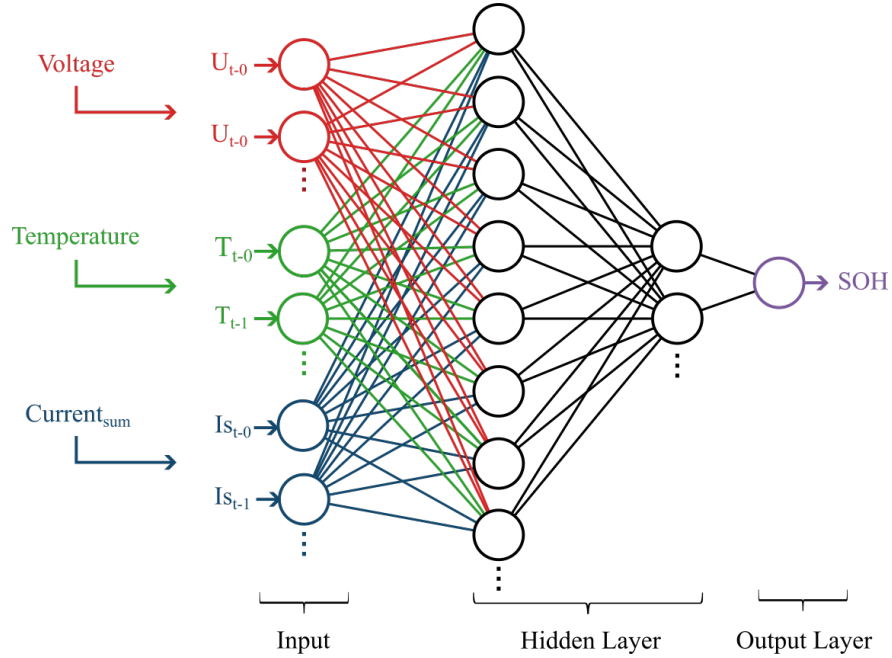


Figure 5.4: Study-specific MLP structure for SOH estimation. The flattened input layer contains voltage, temperature, and cumulative-current values from the lag sequence. These feature–time entries are fully connected to the dense hidden layers, and a single linear output provides the SOH estimate. The displayed node counts are schematic, while the tuned layer widths are reported in Table 5.1.